Depth	Width	Accuracy (%) (standard training)	Accuracy (%) (PC regularization)
CIFAR-10 subsampled (6% of the full dataset):			
56	1	77.8 ± 0.429	**79.2 ± 0.188**
26	8	80.1 ± 0.354	**81.1 ± 0.185**
26	10	80.3 ± 0.306	**81.2 ± 0.194**
38	8	80.2 ± 0.362	**80.9 ± 0.264**
38	10	80.3 ± 0.412	**81.4 ± 0.350**
CIFAR-10:			
110	1	94.3 ± 0.078	**94.4 ± 0.063**
164	1	94.4 ± 0.075	**94.5 ± 0.063**
26	10	95.8 ± 0.087	**96.0 ± 0.051**
38	8	95.7 ± 0.091	**95.8 ± 0.080**
38	10	95.7 ± 0.157	**95.9 ± 0.067**
CIFAR-100:			
218	1	74.1 ± 0.310	**75.1 ± 0.132**
224	1	74.0 ± 0.350	**75.2 ± 0.131**
38	8	79.8 ± 0.149	**80.6 ± 0.306**
38	10	80.5 ± 0.174	**81.1 ± 0.241**

Table G.1: Comparison of performance of large capacity models on CIFAR-10 and CIFAR-100, with and without principal component regularization. We observe that our proposed principal component regularizer consistently yields accuracy improvements for large capacity models that contain the block structure. The performance gains are particularly significant in the case of CIFAR-100 and subsampled CIFAR-10 datasets.

H Effect of Batch Size on the Block Structure

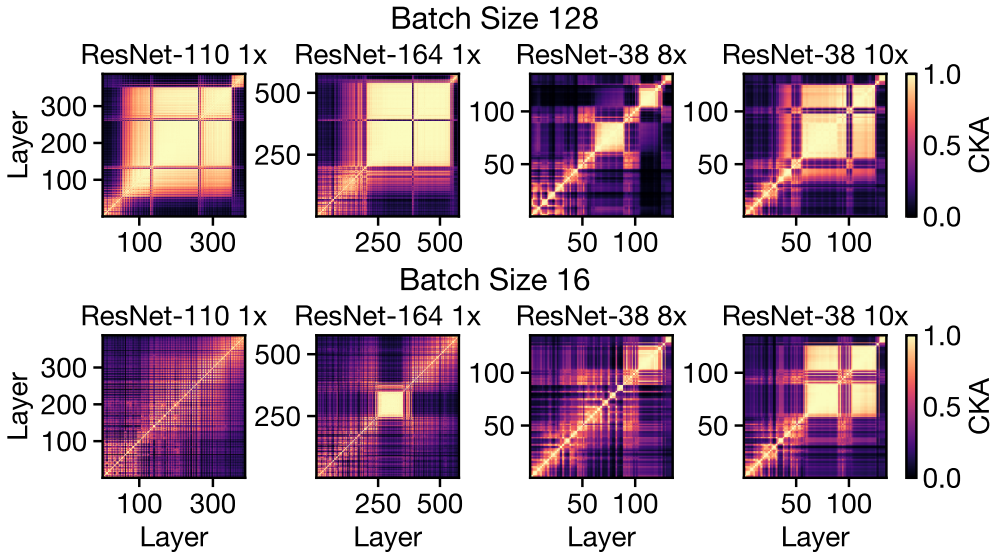

Figure H.1: Using very small batch sizes during training reduces the appearance of the block structure. The top row shows the block structure effect in a range of very deep and wide networks, trained with standard batch size = 128. We experiment with a drastically smaller batch size of 16 (bottom row) and find that the block structure is now highly reduced, especially in deep models.

I Impact of Transfer Learning and Shake-Shake Regularization on Similarity of Layers Inside the Block Structure

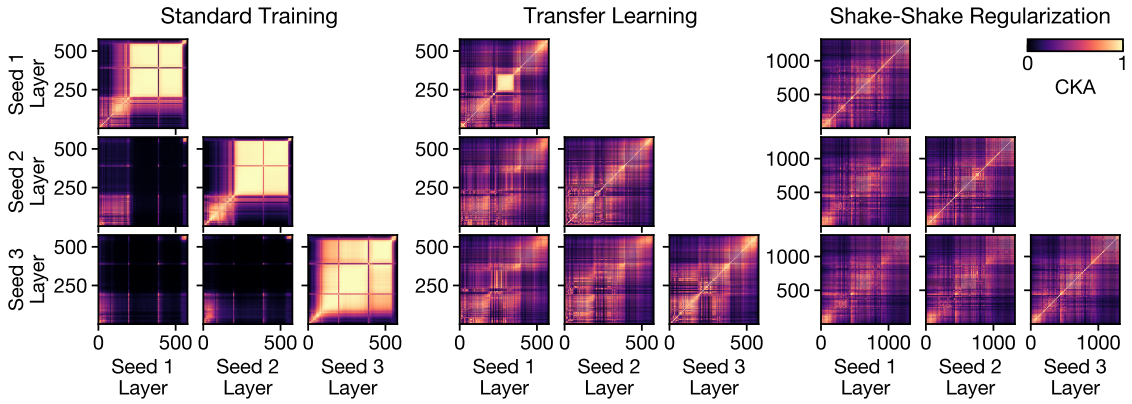

Figure I.1: Training with transfer learning and Shake-Shake regularization yields models that are more similar across different training runs. Each group of plots shows CKA between layers of models with the same architecture but different initializations (off the diagonal) or within a single model (on the diagonal). In the standard training case, representations across models are highly dissimilar, especially in the block structure region. In contrast, when we use transfer learning and Shake-Shake regularization, comparisons across seeds show more similarity in corresponding layers. The same observation can be made for models trained with principal component regularization (see Appendix Figure I.2).

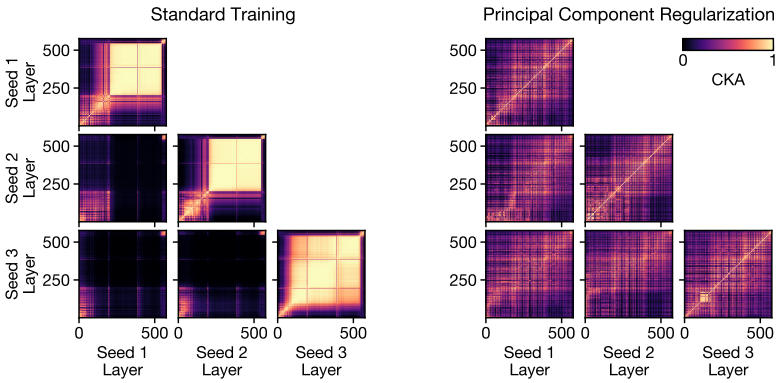

Figure I.2: Training with principal component regularization yields models that are more similar across different training runs. Each group of plots shows CKA between layers of models with the same architecture but different initializations (off the diagonal) or within a single model (on the diagonal). Similar to the observation made in Figure I.1, while representations across models are highly dissimilar in the standard training case, models trained with principal component regularization from different random initializations show more representational similarity in corresponding layers.